KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

Department of Artificial Intelligence and Data Science

EXPLORATORY DATA ANALYSIS AND VISUALIZATION

PROJECT REPORT

CAMPAIGNPULSE

Exploring Customer Response to Bank Term-Deposit Campaigns with EDA and Deep Learning

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Dataset: UCI Bank Marketing - bank-additional-full.csv

Abstract

This project applies exploratory data analysis (EDA), statistical visualization, and Deep Learning to study responses to a bank term-deposit marketing campaign. It uses the bank-additional-full variant of the UCI Bank Marketing dataset, with 41,188 records, 20 input attributes, and a binary subscription target. The workflow includes data-quality checks, treatment of categorical values, outlier screening, descriptive statistics, and eight EDA visualizations plus an interactive plot. Numerical inputs are imputed and standardized, categorical inputs are imputed and one-hot encoded, and duration is excluded from the predictive inputs to avoid post-call target leakage. Two Multilayer Perceptron (MLP) configurations are compared using validation data. The selected MLP_A achieved 84.37% test accuracy and a ROC-AUC of 0.8009. However, subscriber precision was 38.50%, indicating substantial false-positive predictions. The findings highlight target imbalance and historical associations that require careful interpretation.

Keywords: Exploratory Data Analysis, Bank Marketing, Data Visualization, MLP, Deep Learning, Subscription Prediction

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1. Introduction and Objective

Banks use direct-marketing campaigns to inform potential customers about financial products such as term deposits. Historical campaign records can be explored to understand how subscription outcomes vary with client characteristics, contact timing, campaign history, and economic context. CampaignPulse applies EDA, statistical analysis, visualization, and a Deep Learning classifier to the Bank Marketing dataset published by the UCI Machine Learning Repository [1].

1.1 Problem Statement

The project investigates which client and campaign attributes are associated with subscription to a term deposit. The target variable y indicates whether a contacted client subscribed (yes) or did not subscribe (no). The study is descriptive and predictive; it does not establish that any individual feature causes subscription.

1.2 Objectives

- Inspect the dataset structure, feature types, target values, and basic data quality.
- Explore individual numerical distributions, category-wise subscription rates, and relationships among numerical features.
- Create at least five meaningful visualizations with an explanation of purpose and insight.
- Document the seven visualization stages: acquire, parse, filter, mine, represent, refine, and interact [3].
- Prepare training, validation, and test partitions and train Deep Learning MLP configurations.
- Evaluate the selected model with accuracy, precision, recall, F1-score, ROC-AUC, a confusion matrix, and ROC curve.
- Use permutation importance to study predictive feature reliance without interpreting it as causation.

2. Dataset Description

The selected file is bank-additional-full.csv from the UCI Bank Marketing collection. It contains historical direct-marketing records from a Portuguese banking institution. The full additional variant contains 41,188 records with 20 input attributes and target y, and the observations are ordered by date from May 2008 to November 2010 [1].

Dataset item	Description
Dataset title	Bank Marketing - bank-additional-full.csv
Official source	UCI Machine Learning Repository
Direct URL	https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing
Dataset DOI	10.24432/C5K306
Records	41,188
Input attributes	20
Target column	y: yes / no term-deposit subscription
Task	Binary classification
File format	Semicolon-separated CSV
License	Creative Commons Attribution 4.0 (CC BY 4.0), according to UCI [1]

2.1 Main Feature Groups

Feature group	Examples	Purpose in analysis
Client profile	age, job, marital, education	Describe who was contacted.
Credit and household indicators	default, housing, loan	Represent available customer information.
Contact and campaign history	contact, month, day_of_week, campaign, pdays, previous, poutcome	Describe the current contact and previous campaign interactions.
Call duration	duration	Used for exploratory analysis only; excluded from predictive inputs.
Economic indicators	emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed	Represent macroeconomic context.
Target	y	Whether the client subscribed to the term deposit.

2.2 Observed Target Distribution

The executed target-distribution chart shows 36,548 non-subscriptions (88.73%) and 4,640 subscriptions (11.27%). This imbalance is a central consideration in model development and evaluation: a model that predicts the majority class for every record would achieve 88.73% accuracy but would identify no actual subscribers.

3. EDA and Preprocessing

3.1 Data Understanding and Quality Checks

The analysis inspected dimensions, column names, data types, numerical summaries, category frequencies, missing-value counts, and exact duplicate-row counts. The categorical label unknown is present in the source and was preserved as an explicit category instead of being automatically treated as a missing value. Potential outliers were checked with distribution plots and the interquartile-range rule; observations were not removed solely because they were extreme, since long calls and larger contact counts can be genuine records.

3.2 Preprocessing Decisions

- Target encoding: y = no was mapped to 0 and y = yes was mapped to 1.
- Leakage prevention: duration was retained for EDA but excluded from model features because the current call duration is known only after the call is completed.
- Numerical processing: numerical input fields were median-imputed and standardized using StandardScaler.
- Categorical processing: categorical fields were imputed using the most-frequent value and converted using one-hot encoding.
- Fit discipline: the preprocessing pipeline was fitted on training data only and then applied to validation and test data.
- Class imbalance: balanced class weights were used during MLP training.
- Outlier display: duration and contact-count charts were capped at the 99th percentile for visual readability; the underlying records were retained.

3.3 Train / Validation / Test Split

A stratified 70:15:15 split was used to preserve the class distribution across partitions. The screenshot of the model evaluation reports a test support of 6,179 records, including 5,483 non-subscribers and 696 subscribers.

Partition	Approximate records	Use
Training	28,831	Fit model parameters and class weighting.
Validation	6,178	Compare configurations and select a model.
Test	6,179	Final evaluation after model selection.
Total	41,188	Full dataset.

4. Data Visualization (Minimum 5 Visuals)

The following eight static visualizations and one interactive exploration were selected to cover univariate distributions, category-level subscription rates, bivariate and multivariate relationships, and interactive inspection. Each figure states what it shows, why it was created, and the insight supported by the observed output.

Figure 1. Subscription Target Distribution

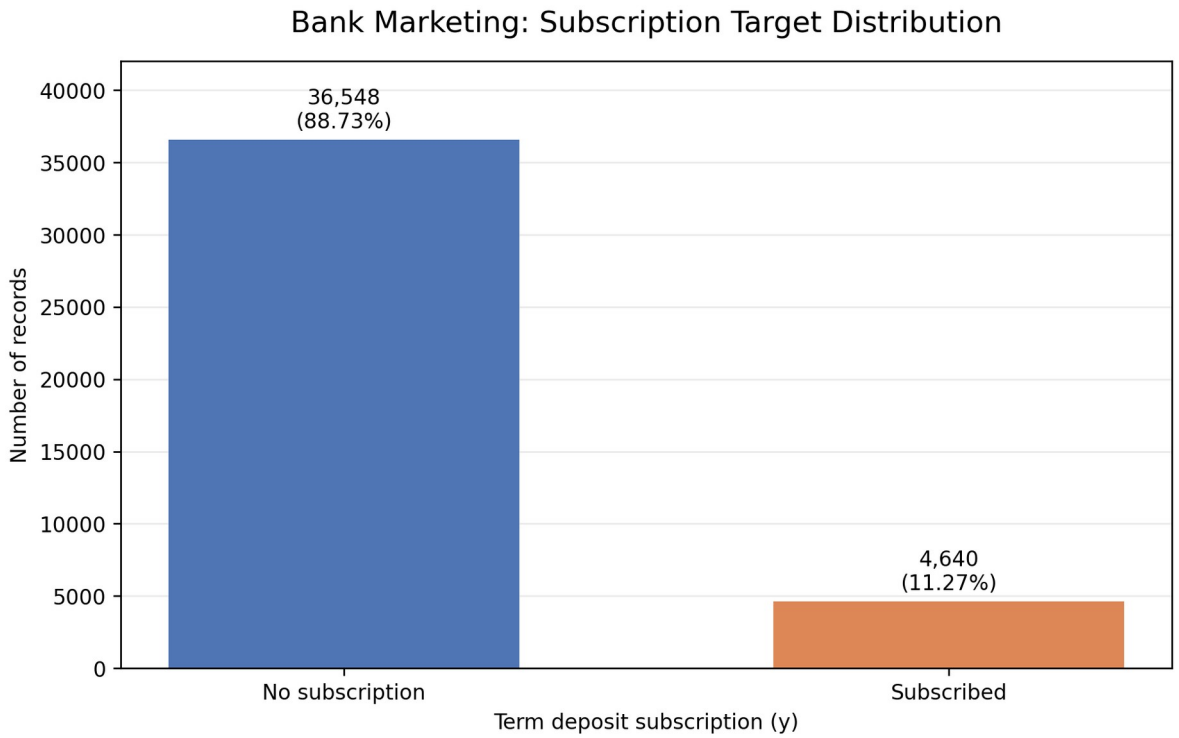


Figure 1: Subscription Target Distribution. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The chart displays the count and percentage of records where y is no or yes.

Why it was created: To assess the class balance before modeling and decide which evaluation measures are needed.

Insight from the output: Non-subscriptions account for 88.73% (36,548 records) and subscriptions for 11.27% (4,640 records). Because the majority class is dominant, accuracy alone can be misleading.

Figure 2. Client Age Distribution

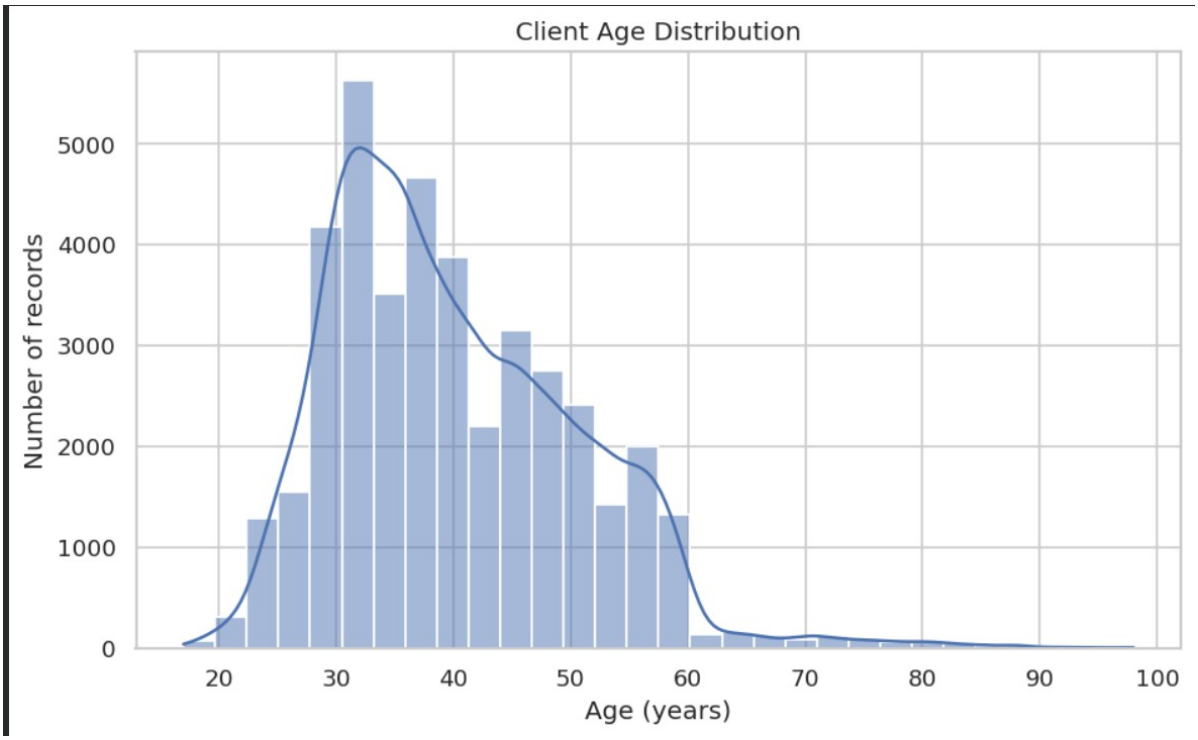


Figure 2: Client Age Distribution. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The histogram with a smooth density curve summarizes the age distribution of contacted clients.

Why it was created: To inspect the spread, concentration, and skew of a numerical customer attribute.

Insight from the output: Most observations are concentrated approximately between the late twenties and late fifties, with a visible peak in the early thirties and a longer tail toward older ages. The chart describes this dataset's contacted clients, not the bank's entire customer population.

Figure 3. Last-Call Duration Distribution

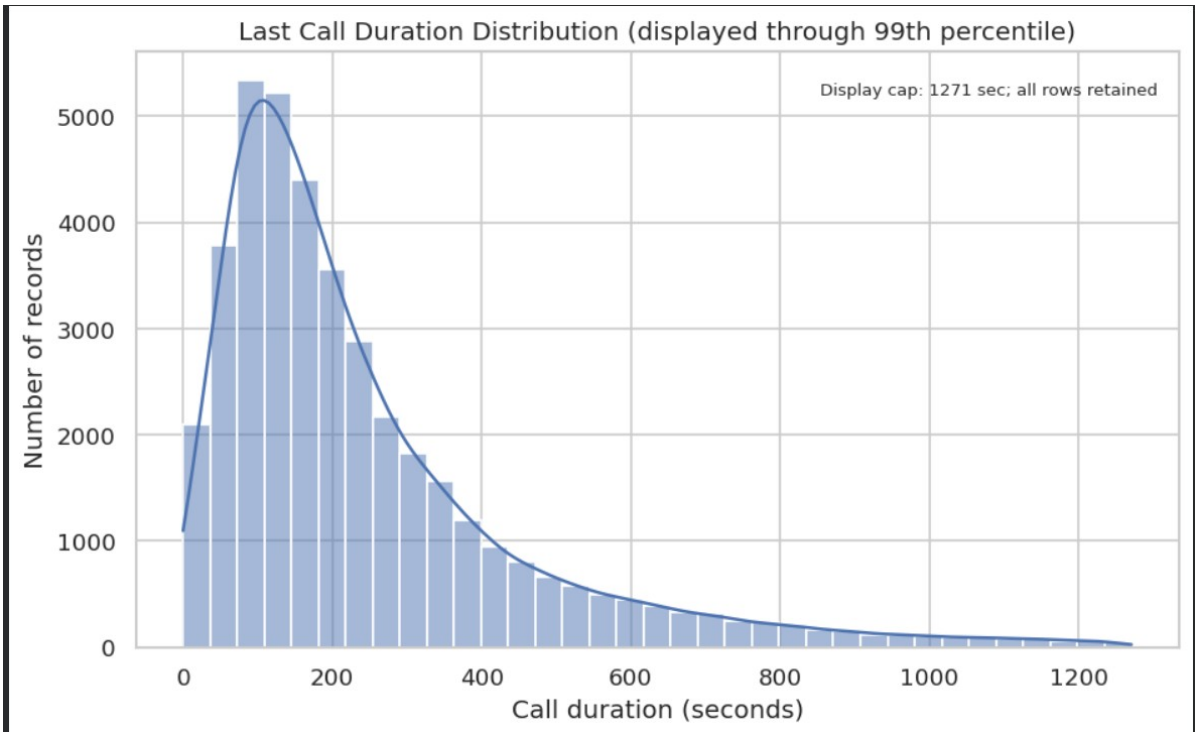


Figure 3: Last-Call Duration Distribution. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The histogram shows the distribution of duration in seconds; the displayed range is capped at the 99th percentile (1,271 seconds).

Why it was created: To understand the strongly skewed call-duration variable without allowing a few long calls to dominate the view.

Insight from the output: The distribution is right-skewed: shorter calls are common and the frequency tapers as call duration increases. All underlying records were retained. Duration is excluded from the prediction features to avoid post-call target leakage.

Figure 4. Number of Contacts in the Current Campaign

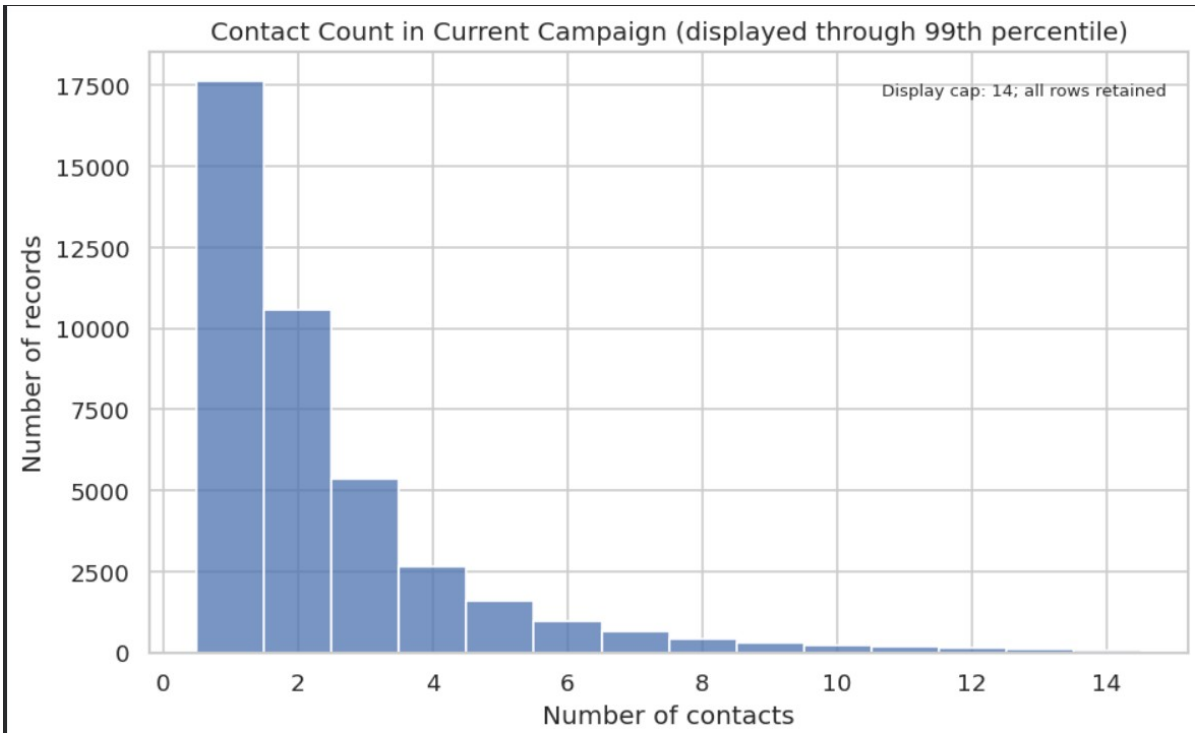


Figure 4: Number of Contacts in the Current Campaign. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The histogram shows the number of contacts recorded for a client in the current campaign, with the displayed range capped at 14 contacts.

Why it was created: To examine how repeated outreach is distributed across records and make the high-frequency tail visible without discarding records.

Insight from the output: The distribution is concentrated at one to a few contacts; records become less frequent as the contact count increases. This is descriptive and does not show that additional contacts cause a subscription or non-subscription.

Figure 5. Subscription Rate by Job Category

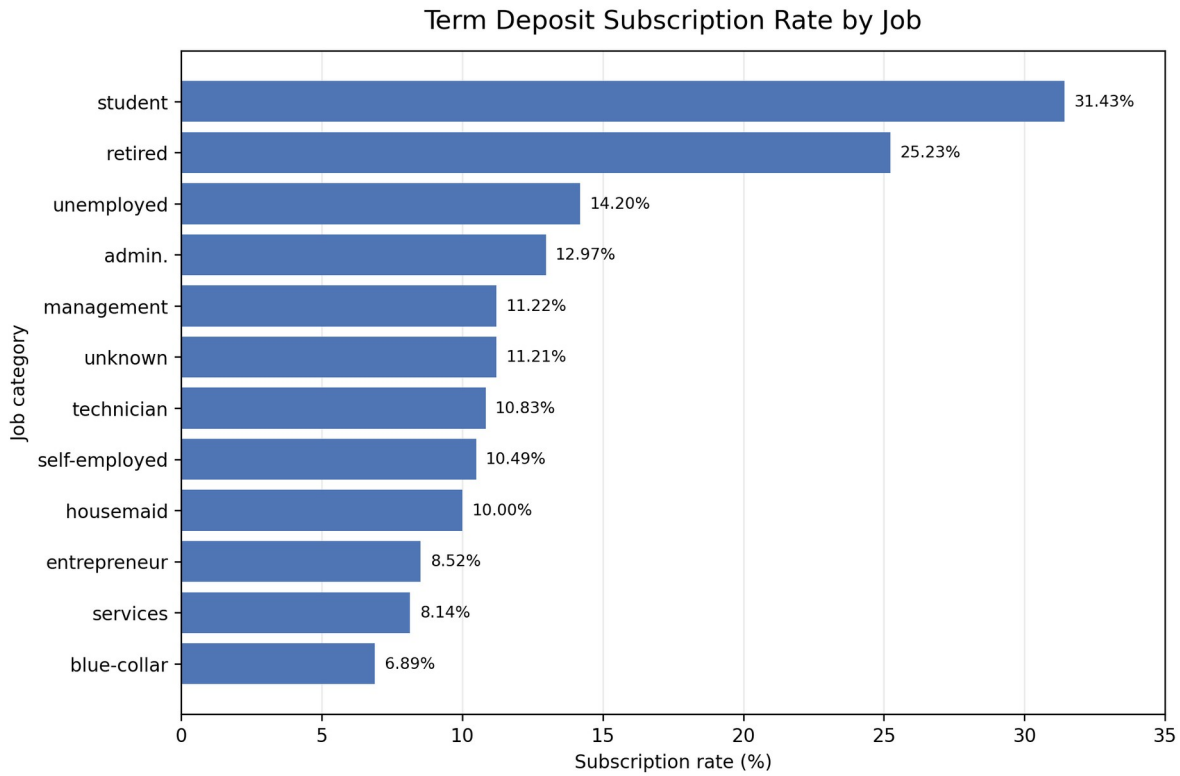


Figure 5: Subscription Rate by Job Category. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The horizontal bar chart compares the within-category subscription rate for each job group.

Why it was created: To compare response rates across categories of a client characteristic rather than comparing raw counts alone.

Insight from the output: Students have the highest shown subscription rate (31.43%), followed by retired clients (25.23%). Blue-collar clients have the lowest shown rate (6.89%). The category rates are descriptive; categories may differ in size, and other variables may explain part of the difference.

Figure 6. Subscription Rate by Contact Month

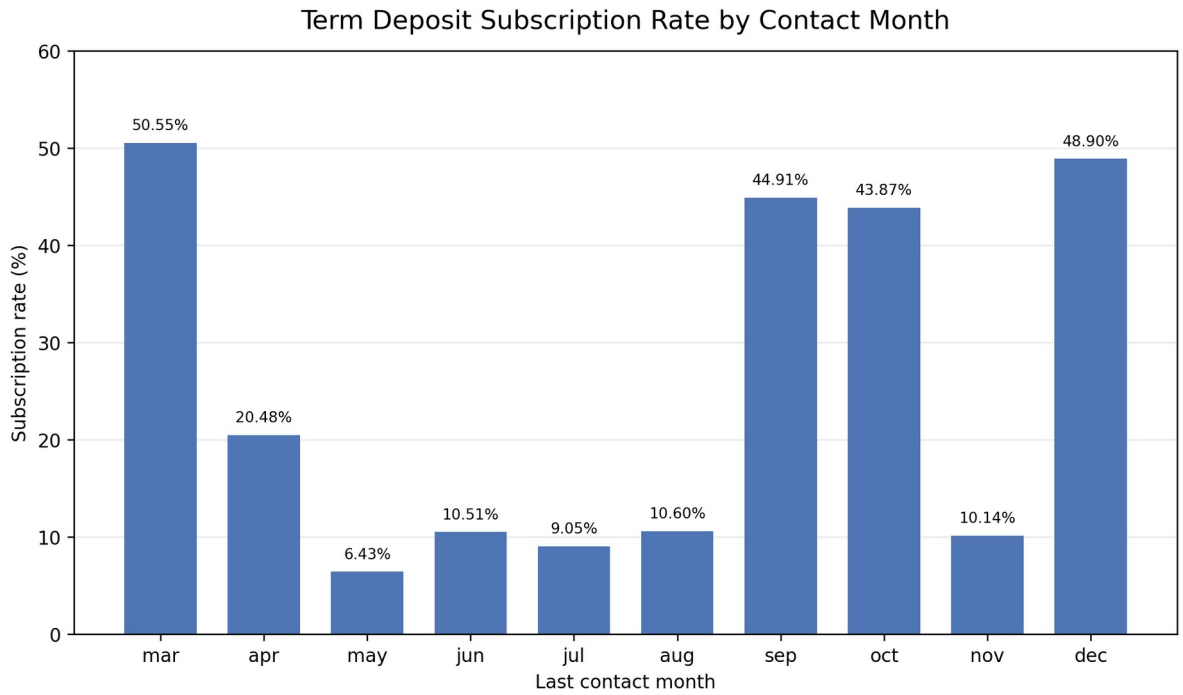


Figure 6: Subscription Rate by Contact Month. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The bar chart shows the subscription percentage for each month present in the executed output.

Why it was created: To explore variation by campaign timing and identify months that warrant closer examination.

Insight from the output: The highest displayed rates are March (50.55%), December (48.90%), September (44.91%), and October (43.87%). May is lowest among the shown months (6.43%). These historical differences should not be interpreted as causal effects of month; campaign mix and period-specific context may differ.

Figure 7. Correlation Heatmap of Numerical Features

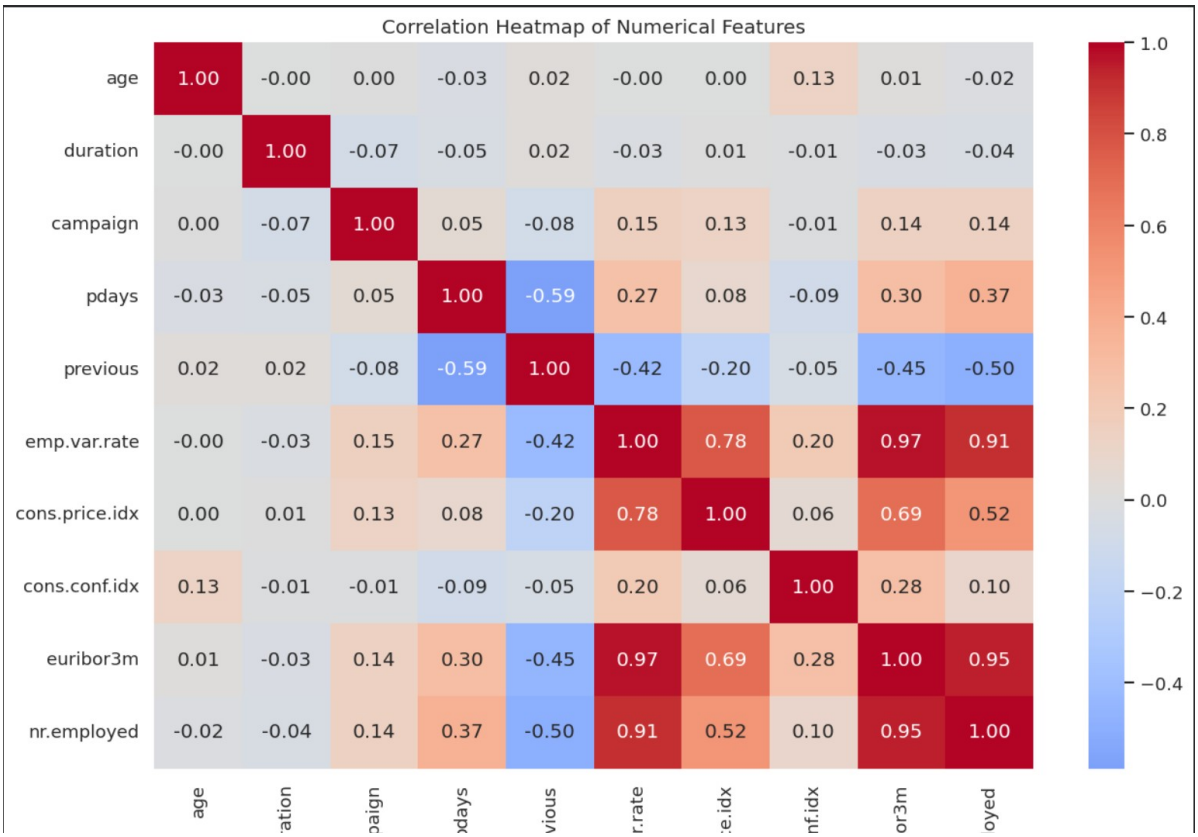


Figure 7: Correlation Heatmap of Numerical Features. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The heatmap displays pairwise Pearson correlations among numerical and economic variables.

Why it was created: To identify strongly associated numerical inputs, potential redundancy, and relationships that need careful interpretation.

Insight from the output: The economic indicators show strong positive associations: emp.var.rate with euribor3m is about 0.97, euribor3m with nr.employed about 0.95, and emp.var.rate with nr.employed about 0.91. These related variables may carry overlapping information. Correlation does not imply causation.

Figure 8. Subscription Rate by Previous Campaign Outcome

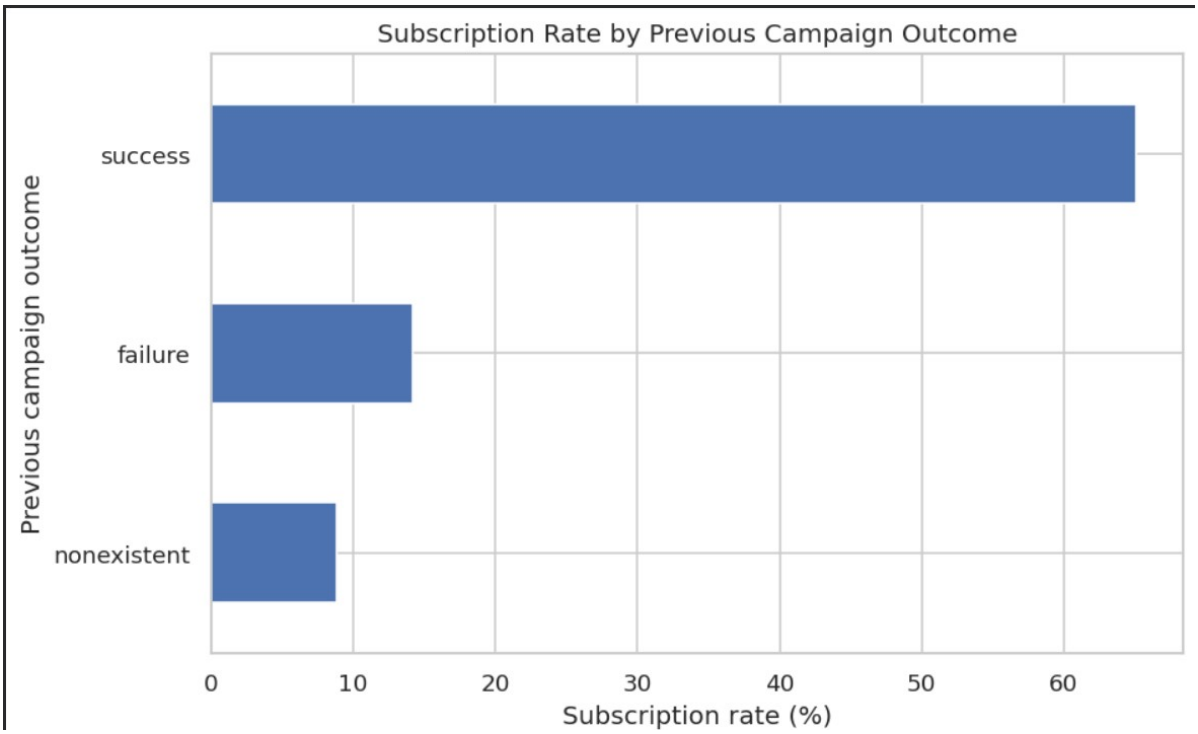


Figure 8: Subscription Rate by Previous Campaign Outcome. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The horizontal bar chart compares subscription rates for previous campaign outcomes: success, failure, and nonexistent.

Why it was created: To examine whether recorded history from a previous campaign is associated with response in the current campaign.

Insight from the output: The success category has a much higher subscription rate than failure or nonexistent outcomes in the displayed chart. This indicates that prior campaign history is informative, but it does not guarantee a future subscription.

Figure 9. Interactive Exploration: Age versus Campaign Contacts

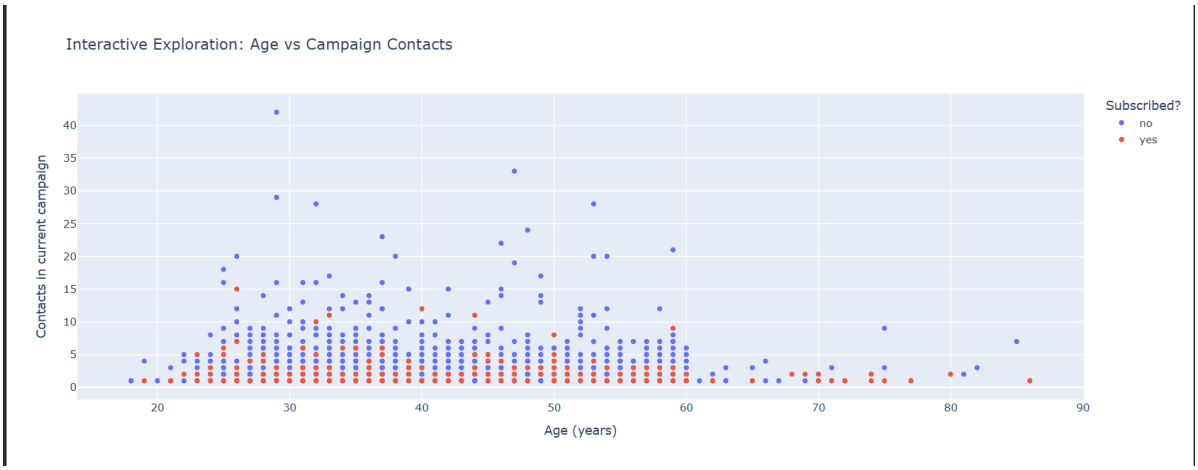


Figure 9: Interactive Exploration: Age versus Campaign Contacts. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The Plotly screenshot explores client age against the number of contacts in the current campaign, colored by the target outcome.

Why it was created: To demonstrate the Interact stage, allowing a user in the notebook to hover over sampled points and inspect feature values.

Insight from the output: Most plotted records have a small number of contacts and fall roughly between ages 25 and 60. Subscriptions and non-subscriptions appear across multiple age/contact combinations, showing that no single pair of variables completely separates the classes. The interactive graph uses a reproducible sample for responsiveness.

4.1 The Seven Stages of Visualization

The project maps its workflow to the seven-stage visualization framework described by Fry [3]. The stages are iterative rather than a rigid one-way sequence.

Stage	Implementation in CampaignPulse
1. Acquire	Download bank-additional-full.csv from the UCI Machine Learning Repository.
2. Parse	Read the semicolon-separated CSV and inspect column names, types, and target labels.
3. Filter	Inspect data quality and select appropriate analytical inputs; retain legitimate observations and exclude duration from predictors for leakage prevention.
4. Mine	Compute distribution summaries, category subscription rates, and numerical correlations.
5. Represent	Use histograms, bar charts, a correlation heatmap, and a scatter plot.
6. Refine	Improve titles, labels, displayed ranges, legends, and figure captions.
7. Interact	Use a Plotly scatter plot with hover data to explore age and campaign contacts.

5. Deep Learning Model

A feed-forward Multilayer Perceptron (MLP) was implemented with TensorFlow/Keras. The network accepts transformed numerical and categorical features and outputs a single probability for subscription using a sigmoid output neuron. The Sequential API is suitable for stacking the dense layers used here [4].

5.1 Selected Architecture

Layer	Configuration	Purpose
Input	Preprocessed model features; duration excluded	Receives numeric and one-hot encoded client/campaign attributes.
Dense 1	64 units, ReLU (MLP_A)	Learns nonlinear combinations of features.
Dropout 1	0.25	Regularization during training.
Dense 2	32 units, ReLU (MLP_A)	Learns a compact intermediate representation.
Dropout 2	0.25	Regularization during training.
Output	1 unit, sigmoid	Returns a probability for y = yes.

5.2 Training Parameters and Hyperparameter Comparison

Setting	MLP_A (selected)	MLP_B
Hidden units	(64, 32)	(32, 16)
Dropout rate	0.25	0.30
Learning rate	0.001	0.0005
Optimizer	Adam	Adam
Loss	Binary cross-entropy	Binary cross-entropy
Batch size	256	256
Maximum epochs	30	30
Early stopping	Monitor validation AUC; patience 4; restore best weights	Same protocol
Class imbalance	Balanced class weights	Balanced class weights

The executed output identifies MLP_A as the selected model based on validation ROC-AUC. The test partition was reserved for final evaluation after selection. Exact per-configuration validation AUC values were not visible in the submitted output screenshots, so no unobserved comparison scores are reproduced in this report.

Figure 10. Training and Validation Loss / Accuracy

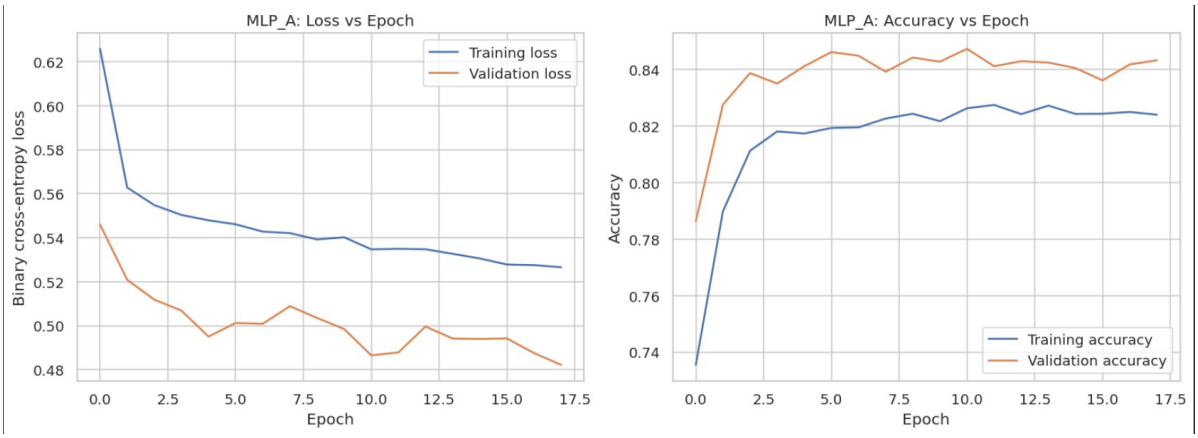


Figure 10: Training and Validation Loss / Accuracy. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The two panels show the selected MLP_A training and validation loss and accuracy by epoch.

Why it was created: To assess the learning process and check for signs that training performance continues to improve while validation performance deteriorates.

Insight from the output: The plotted training loss decreases from roughly 0.63 to 0.53 and validation loss from about 0.55 to 0.48. Validation accuracy remains around 0.84 while training accuracy is around 0.82 in the later epochs. The displayed curves do not show a pronounced widening loss gap, although evaluation metrics on held-out data remain decisive.

6. Result Visualization and Interpretation

6.1 Final Test Metrics

The final evaluation used a test set of 6,179 records. Metrics below are based on the supplied classification report and confusion matrix. Precision, recall, and F1-score are reported for the positive class (subscribed) as well as the class-specific summary.

Metric	Test value	Interpretation
Accuracy	84.37%	Correct predictions divided by all test records.
Precision (subscribed)	38.50%	Of records predicted as subscribed, the proportion that actually subscribed.
Recall (subscribed)	64.94%	Of actual subscribers, the proportion identified by the model.
F1-score (subscribed)	48.34%	Harmonic mean of subscriber precision and recall.
ROC-AUC	0.8009	Ability to rank subscribers above non-subscribers across thresholds.

6.2 Class-wise Results

Class	Precision	Recall	F1-score	Support
No subscription	0.95	0.87	0.91	5,483
Subscribed	0.39	0.65	0.48	696
Accuracy	-	-	0.84	6,179
Macro average	0.67	0.76	0.70	6,179
Weighted average	0.89	0.84	0.86	6,179

Interpretation: subscriber recall is substantially higher than zero, but subscriber precision is low. The model identifies many actual subscribers, yet it also labels many non-subscribers as potential subscribers. Accuracy (84.37%) is below the 88.73% majority-class baseline; therefore, performance should be discussed in terms of the intended goal of detecting subscribers rather than accuracy alone.

Figure 11. Confusion Matrix - Bank Term Deposit Subscription

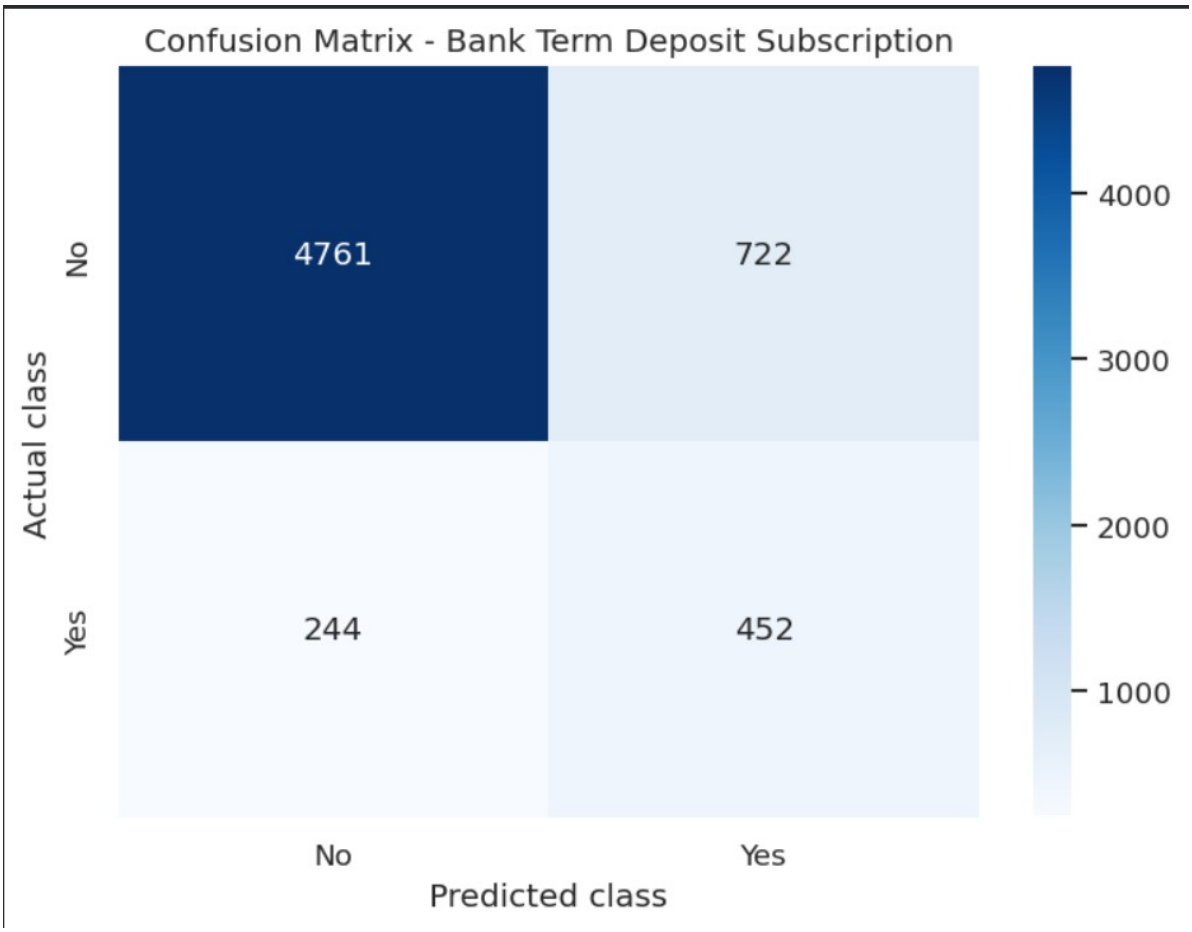


Figure 11: Confusion Matrix - Bank Term Deposit Subscription. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The matrix compares actual and predicted outcomes. Rows are actual classes and columns are predicted classes.

Why it was created: To distinguish correct predictions from the two types of classification error.

Insight from the output: The model correctly predicted 4,761 non-subscribers and 452 subscribers. It produced 722 false positives (non-subscribers predicted as subscribers) and 244 false negatives (subscribers predicted as non-subscribers). The 244 false negatives correspond to the 64.94% recall for the subscribed class.

Figure 12. ROC Curve - Bank Marketing Subscription

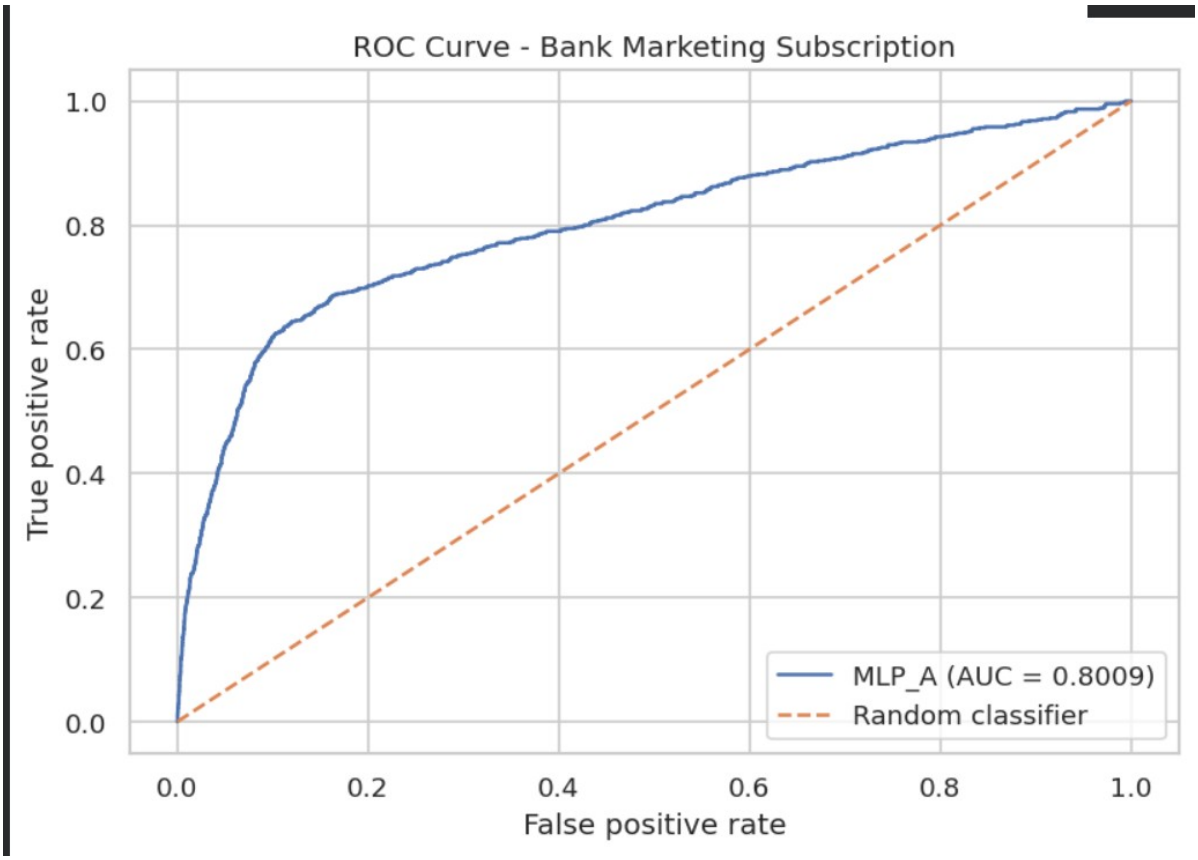


Figure 12: ROC Curve - Bank Marketing Subscription. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The ROC curve plots the true-positive rate against the false-positive rate across classification thresholds.

Why it was created: To evaluate how well model scores distinguish the two classes independently of one fixed decision threshold.

Insight from the output: The reported test ROC-AUC is 0.8009, indicating useful ranking discrimination above random ranking. The curve does not by itself determine an operating threshold; threshold selection should consider the cost of false positives and missed subscribers.

Figure 13. Permutation Feature Importance

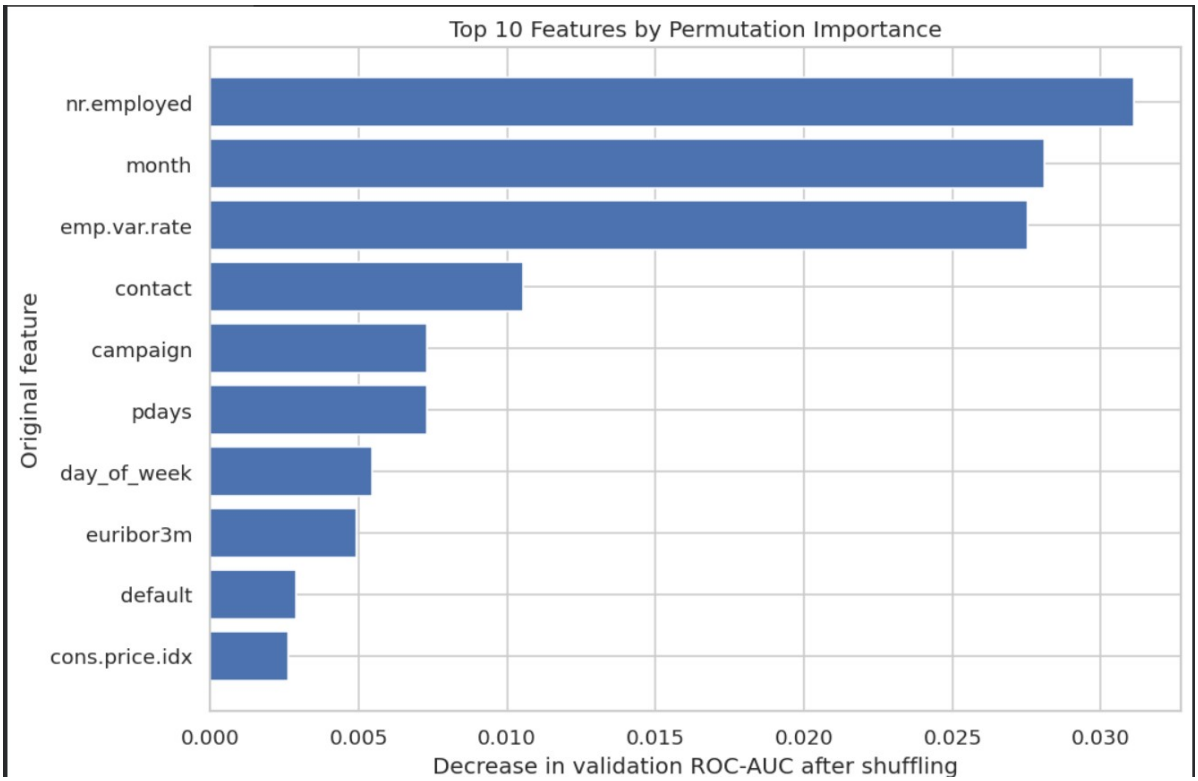


Figure 13: Permutation Feature Importance. Source: CampaignPulse output screenshots; selected rate charts were redrawn from the displayed values for legibility.

What it shows: The chart ranks original input features by the decrease in validation ROC-AUC observed when each feature is shuffled.

Why it was created: To investigate which inputs the selected MLP relies on for predictive discrimination.

Insight from the output: The leading features are nr.employed, month, emp.var.rate, contact, campaign, pdays, day_of_week, euribor3m, default, and cons.price.idx. The first three stand out in the chart. Importance measures model reliance, not causation; correlated economic features may share predictive information.

7. Conclusion and Future Scope

7.1 Conclusion

CampaignPulse applied EDA, statistical summaries, data visualization, and a Deep Learning MLP to the UCI Bank Marketing dataset containing 41,188 campaign records. The target distribution is imbalanced: 11.27% of contacts resulted in term-deposit subscription, while 88.73% did not. The exploratory visualizations showed that observed subscription rates vary across job categories, contact months, and previous campaign outcomes. The numerical correlation heatmap also showed strong relationships among economic indicators.

The selected MLP_A achieved 84.37% test accuracy, 38.50% precision for the subscribed class, 64.94% recall, 48.34% F1-score, and ROC-AUC 0.8009. The confusion matrix indicates 452 true-positive subscriber predictions, 244 missed subscribers, and 722 false-positive predictions. Its accuracy is lower than the 88.73% majority-class baseline, but it identifies approximately 65% of actual subscribers. This trade-off is important: usefulness depends on whether the bank values subscriber detection enough to tolerate the false-positive workload. The model is an exploratory baseline rather than a deployment-ready decision system.

7.2 Future Scope

- Tune the probability threshold using validation data to examine the precision-recall trade-off; keep the test set untouched until the final evaluation.
- Compare the MLP with logistic regression and tree-based models using the same split and metrics.
- Use repeated cross-validation or chronological evaluation to understand result stability over time.
- Investigate class-imbalance strategies and probability calibration.
- Review subgroup performance and possible fairness impacts before using predictions to prioritize customer contact.
- Avoid post-call predictors such as duration when the objective is to predict subscription before or during a call.

8. Literature Review and References

The UCI Machine Learning Repository provides the official dataset description, target definition, file variants, and recommended citation for Bank Marketing [1]. Moro, Cortez, and Rita studied data-driven prediction of bank telemarketing success and provide the research context for using customer, campaign, and economic attributes in predictive analysis [2]. Fry describes the seven-stage visualization workflow used to structure this project from data acquisition through interaction [3]. The Keras Sequential API documentation supports the implementation of a layered MLP [4], while the scikit-learn documentation describes permutation importance as a method for measuring how a model evaluation score changes when a feature is shuffled [5]. These sources informed dataset selection, visualization planning, neural-network construction, and cautious interpretation of feature importance.

References

[1] S. Moro, P. Rita, and P. Cortez, “Bank Marketing” [Dataset], UCI Machine Learning Repository, 2014. https://doi.org/10.24432/C5K306. Dataset page: https://archive.ics.uci.edu/dataset/222/bank%2Bmarketing.

[2] S. Moro, P. Cortez, and P. Rita, “A data-driven approach to predict the success of bank telemarketing,” Decision Support Systems, vol. 62, pp. 22–31, 2014. https://doi.org/10.1016/j.dss.2014.03.001.

[3] B. Fry, Visualizing Data: Exploring and Explaining Data with the Processing Environment. O’Reilly Media, 2008. https://www.oreilly.com/library/view/visualizing-data/9780596514556/ch01.html.

[4] Keras, “The Sequential model,” Keras API documentation. https://keras.io/api/models/sequential/.

[5] scikit-learn developers, “Permutation feature importance,” scikit-learn documentation. https://scikit-learn.org/stable/modules/permutation_importance.html.

9. Appendix (Code Section)

The full executable code is provided in CampaignPulse_Deep_Learning.ipynb. The representative extracts below document the main workflow: loading the official dataset, preprocessing, defining and training the MLP, and evaluating it. These extracts are included for readability; the notebook is the authoritative, complete implementation.

A.1 Dataset Loading

```
import io, zipfile, urllib.request
import pandas as pd
url = "https://archive.ics.uci.edu/static/public/222/bank+marketing.zip"
with urllib.request.urlopen(url, timeout=60) as response:
   archive_bytes = response.read()
with zipfile.ZipFile(io.BytesIO(archive_bytes)) as archive:
   csv_path = next(n for n in archive.namelist()
               if n.endswith("bank-additional-full.csv"))
   with archive.open(csv_path) as f:
      df = pd.read_csv(f, sep=";")
print(df.shape)  # expected (41188, 21)
```

A.2 Target Encoding, Split, and Preprocessing

```
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
# Duration is excluded from predictors to avoid target leakage.
X = df.drop(columns=["y", "duration"]).copy()
y = df["y"].map({"no": 0, "yes": 1}).astype(int)
X_train, X_tmp, y_train, y_tmp = train_test_split(
   X, y, test_size=0.30, random_state=42, stratify=y)
X_val, X_test, y_val, y_test = train_test_split(
   X_tmp, y_tmp, test_size=0.50, random_state=42, stratify=y_tmp)
num_cols = X_train.select_dtypes(include=["number"]).columns.tolist()
cat_cols = X_train.select_dtypes(exclude=["number"]).columns.tolist()
num_pipe = Pipeline([("imputer", SimpleImputer(strategy="median")),
                     ("scaler", StandardScaler())])
cat_pipe = Pipeline([("imputer", SimpleImputer(strategy="most_frequent")),
                     ("onehot", OneHotEncoder(handle_unknown="ignore"))])
preprocessor = ColumnTransformer([
   ("num", num_pipe, num_cols), ("cat", cat_pipe, cat_cols)])
X_train_p = preprocessor.fit_transform(X_train)
X_val_p = preprocessor.transform(X_val)
X_test_p = preprocessor.transform(X_test)
```

A.3 MLP Architecture and Training

```
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.callbacks import EarlyStopping
model = keras.Sequential([
   layers.Input(shape=(X_train_p.shape[1],)),
   layers.Dense(64, activation="relu"),
   layers.Dropout(0.25),
   layers.Dense(32, activation="relu"),
   layers.Dropout(0.25),
   layers.Dense(1, activation="sigmoid")
])
model.compile(optimizer=keras.optimizers.Adam(learning_rate=0.001),
              loss="binary_crossentropy",
              metrics=["accuracy", keras.metrics.AUC(name="auc")])
stop = EarlyStopping(monitor="val_auc", mode="max", patience=4,
                     restore_best_weights=True)
# Training uses balanced class weights computed from y_train.
history = model.fit(X_train_p, y_train, validation_data=(X_val_p, y_val),
                    batch_size=256, epochs=30, class_weight=class_weight,
                    callbacks=[stop], verbose=1)
```

A.4 Test Evaluation

```
prob = model.predict(X_test_p, verbose=0).ravel()
pred = (prob >= 0.50).astype(int)
print("Accuracy / Precision / Recall / F1:", accuracy_score(y_test, pred), precision_score(y_test, pred, zero_division=0), recall_score(y_test,
pred, zero_division=0), f1_score(y_test, pred, zero_division=0))
print("ROC-AUC:", roc_auc_score(y_test, prob), "Confusion matrix:", confusion_matrix(y_test, pred))
```